Supplementary information

Efficient Synthesis of CO and H_2O_2 via Nanosecond Pulsed CO_2 Bubble Discharge

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1. Voltage and current curves and discharge power at different pulse frequencies

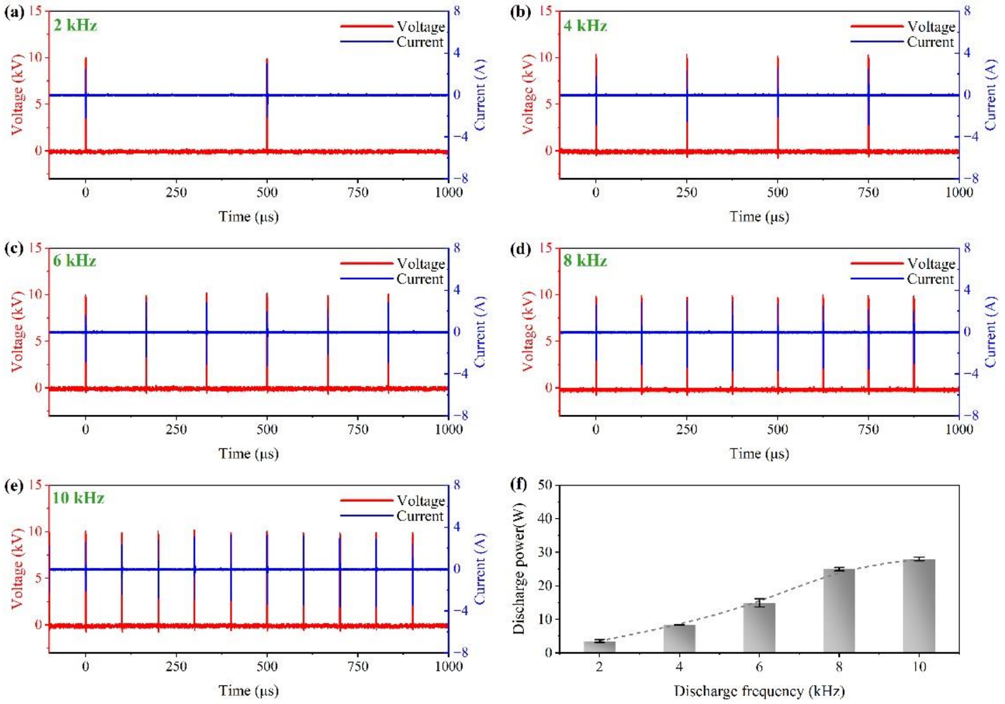

Figure S1. Voltage and current curves and discharge power at different pulse frequencies (Pulse generator settings: voltage: 12 kV, pulse width: 1000 ns, rise time: 50 ns, fall time: 50 ns; gas flow rate: 20 sccm.)

2. Voltage and current curves and discharge power at different pulse widths

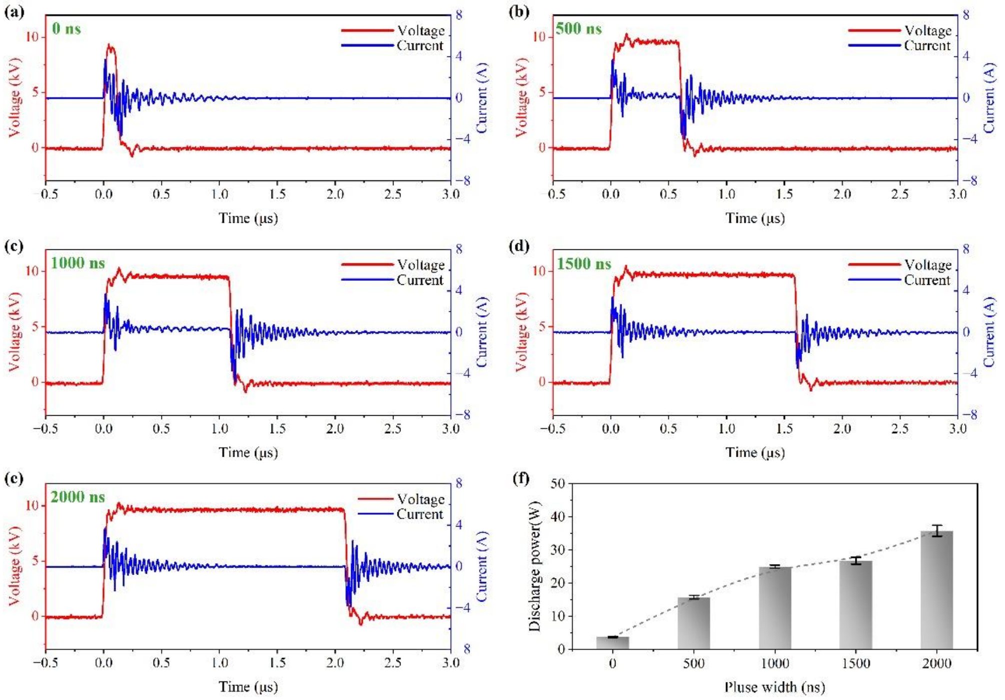

Figure S2. Voltage and current curves and discharge power at different pulse widths. (Pulse generator settings: voltage: 12 kV, pulse frequency: 8 kHz, rise time: 50 ns, fall time: 50 ns; gas flow rate: 20 sccm.)

3. Voltage and current curves and discharge power at different rise times

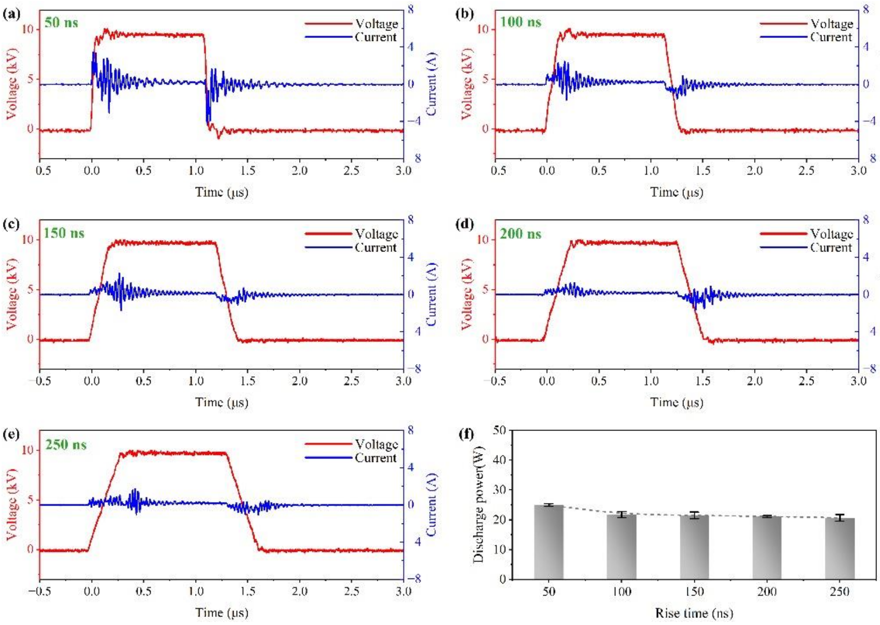

Figure S3. Voltage and current curves and discharge power at different rise times (Pulse generator settings: voltage: 12 kV, pulse frequency: 8 kHz, pulse width: 1000 ns; gas flow rate: 20 sccm.)

4. Voltage and current curves and discharge power at different CO_2 flow rates

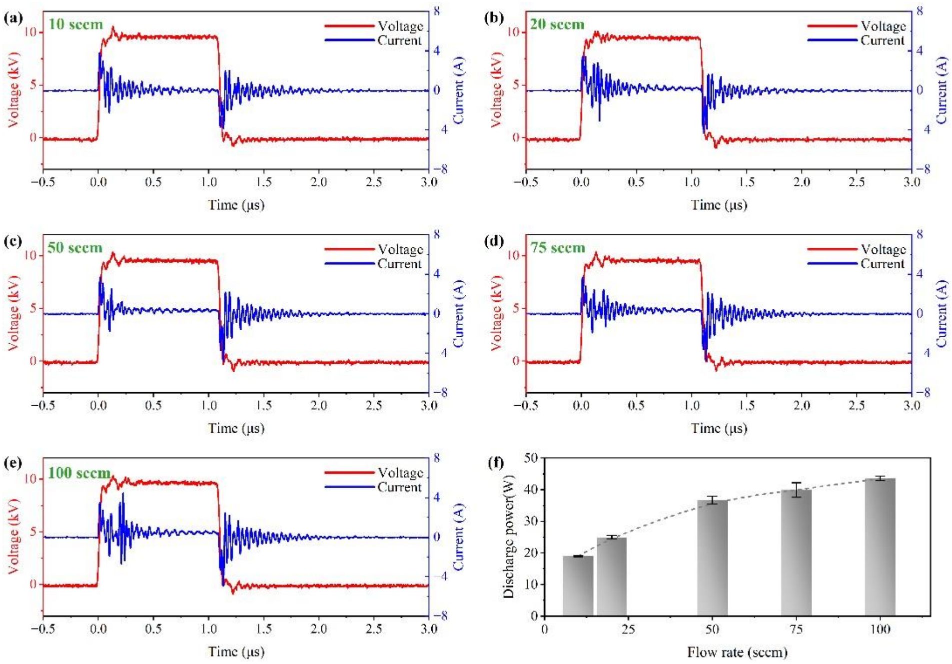

Figure S4. Voltage and current curves and discharge power at different CO_2 flow rates (Pulse generator settings: voltage: 12 kV, pulse frequency: 8 kHz, pulse width: 1000 ns, rise time: 50 ns, fall time: 50 ns.)

5. Reflected power

The energy reflected back to the generator is expressed in terms of reflected power. Defined as the power of the pulse supply minus the power of the plasma discharge. The pulse power supply was connected to an external watt-hour meter to obtain its real-time power consumption.

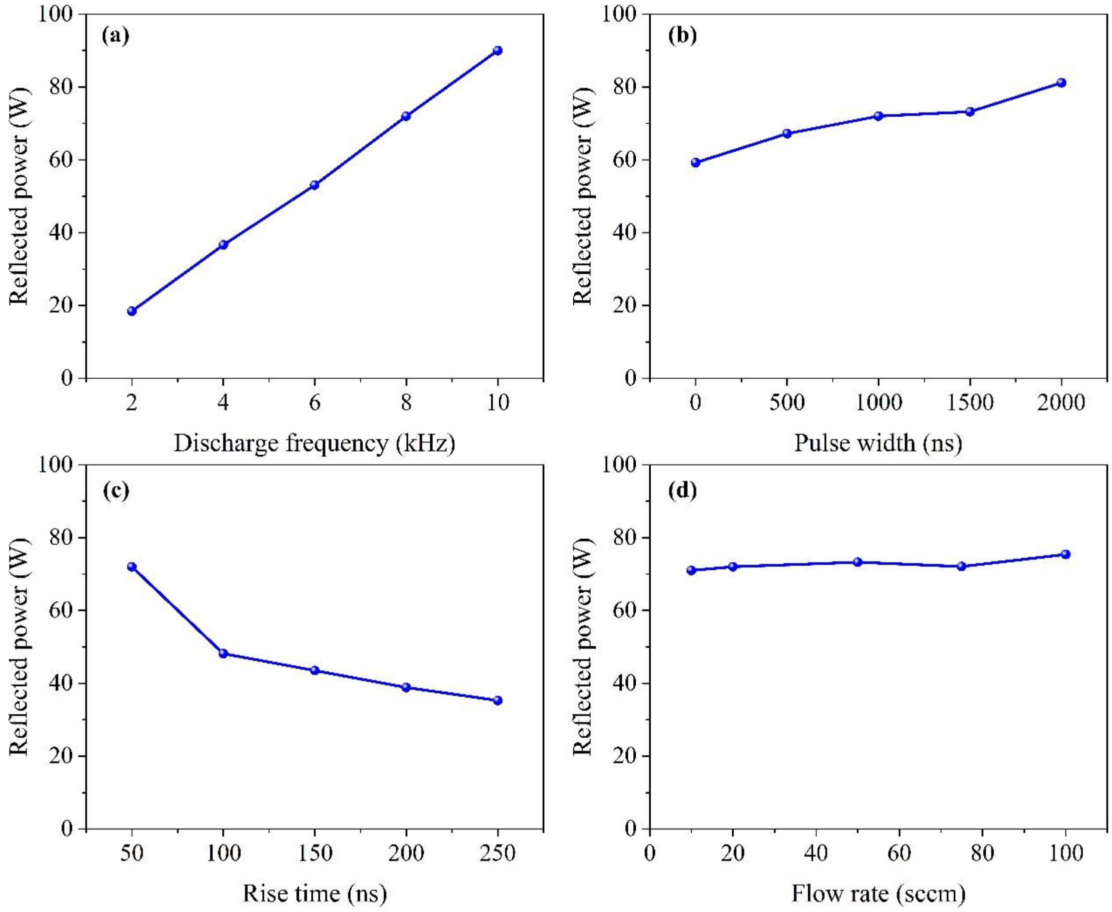

Figure S5. Reflected power under different conditions. (a) different pulse frequencies; (b) different pulse widths; (c) different rise times; (d) different CO_2 flow rates.

The reflected power, as depicted in Figure S5, exhibits an upward trend with the increase of discharge frequency and pulse width, while it shows a downward trend with the rise time increment. When there is a variation in gas flow rate, the reflected power fluctuates within a certain range but can be approximately regarded as constant.

6. Power supply efficiency

Power supply efficiency is defined as the ratio of plasma discharge power to total pulse power supplied. The pulse power supply was connected to an external watt-hour meter to obtain its real-time power consumption.

$$\alpha(\%) = \frac{P_{(discharge)}}{P_{(pulse\ power\ supply)}} \times 100\%$$

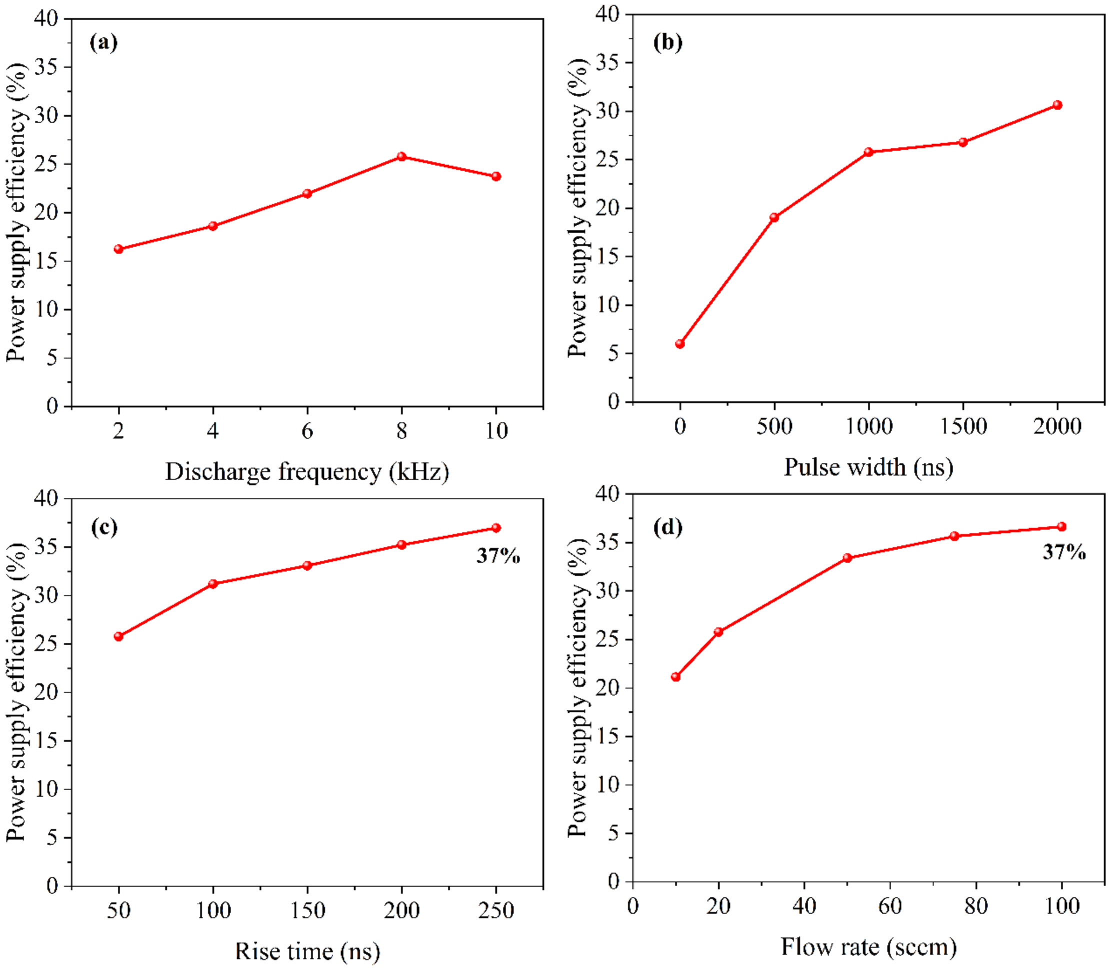

Figure S6. Power supply efficiency under different conditions. (a) different pulse frequencies; (b) different pulse widths; (c) different rise times; (d) different CO_2 flow rates.

As shown in Figure S6, as the frequency increases, the power supply efficiency first increases and then decreases, reaching a maximum value at 8 kHz. With the increase of pulse width, rise time and gas flow rate, the power efficiency gradually increases. Under current conditions, the maximum power efficiency is 37%.

7. Optical emission spectrometry

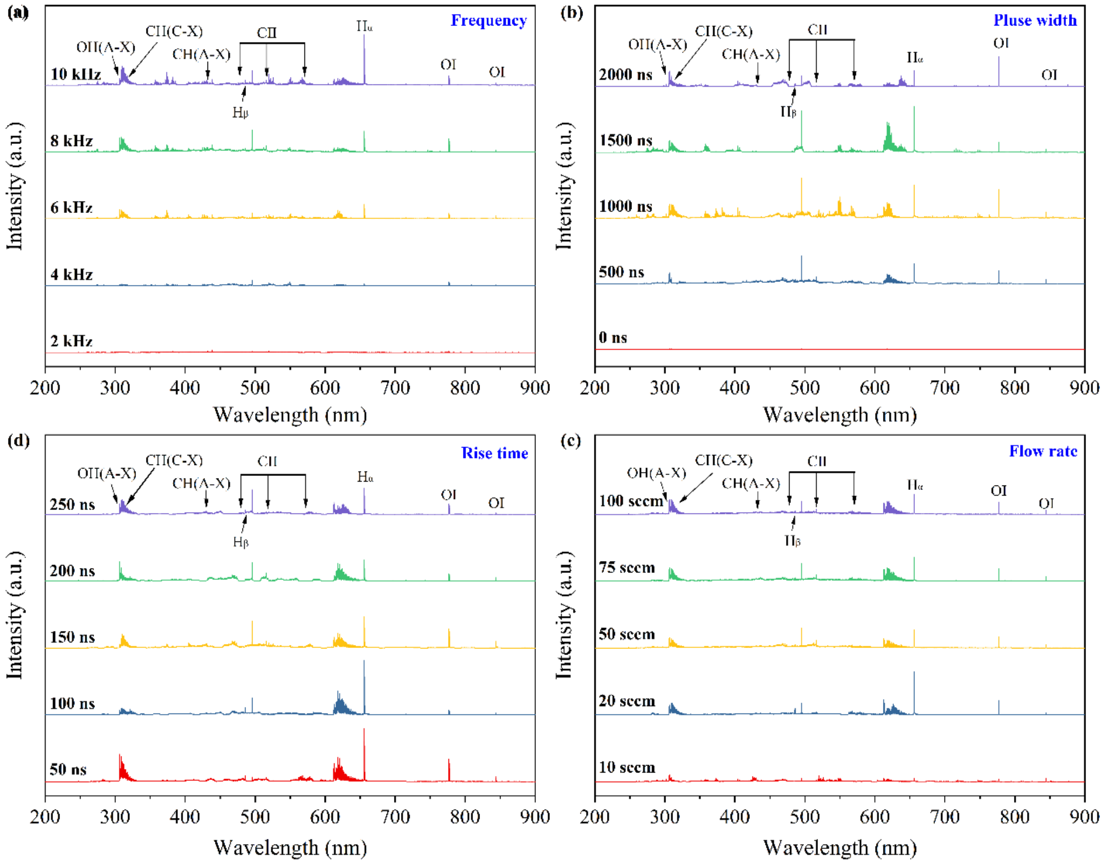

Figure S7. Optical emission spectrometry under different conditions. (a) different pulse frequencies; (b) different pulse widths; (c) different rise times; (d) different CO_2 flow rates.

8. The pH of the solution

At standard atmospheric pressure and 25℃, the solubility of carbon dioxide in water is 0.144g/100 mL. CO is relatively insoluble in water, with a solubility of 0.0026 g/100 mL at 20℃. Compared with CO_2, the solubility of CO is low, so the CO_2 dissolved in water is mainly considered. During the discharge process, some CO_2 dissolves in water to form carbonic acid, so the pH value of the solution after discharge is detected to characterize the CO_2 content dissolved in water.

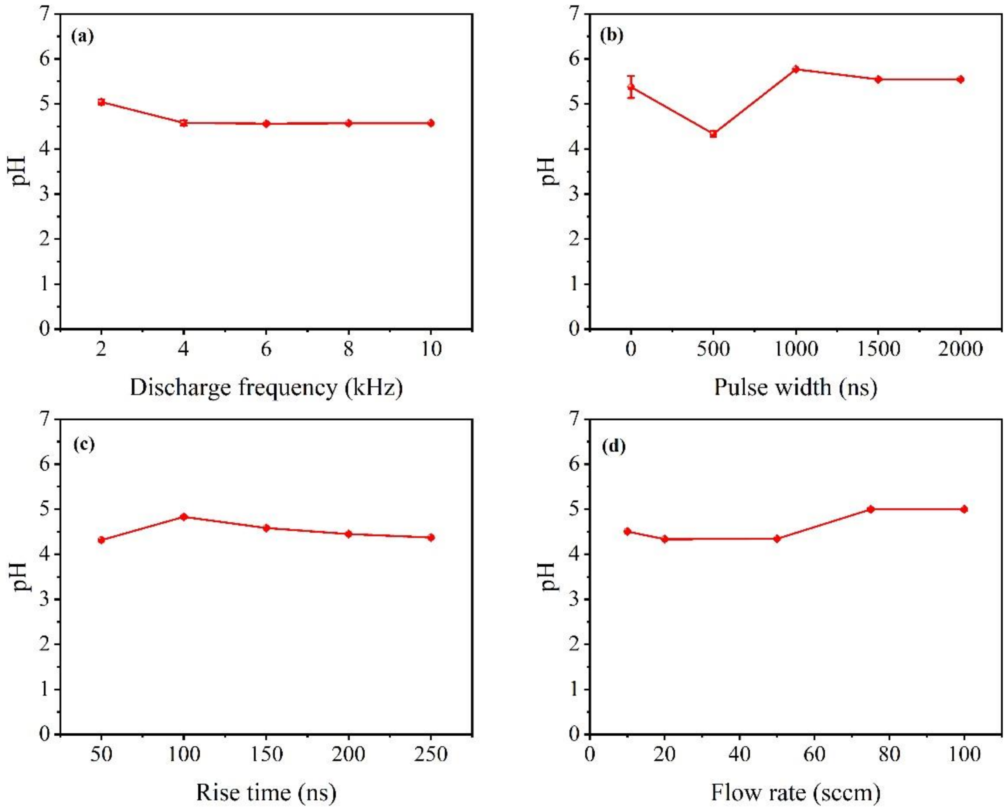

Figure S8. The pH of the solution after discharge under different conditions. (a) different pulse frequencies; (b) different pulse widths; (c) different rise times; (d) different CO_2 flow rates (The discharge time is 2.5 h)

As depicted in Figure S8, the discharge process results in the dissolution of a portion of CO_2 in water, leading to a decrease in the solution's pH value and rendering it weakly acidic. The post-discharge pH value of solution closely approximates that of a saturated CO_2 solution at room temperature and standard atmospheric pressure, falling within the range of 4-5.6.